

RESEARCH INTERESTS

My research lies at the intersection of **generative AI**, **3D computer vision** and **multimodal foundation models**, with a focus on controllable, multi-view, pose- and geometry-aware models for real-world deployment — including virtual try-on, world models for spaces, and on-device generation.

EDUCATION

University of California, Berkeley

Berkeley, CA

Ph.D. in Computer Vision

Aug. 2018 - May 2023

Advisor : [Stella X. Yu](#)

- **Editable Generative AI** : Adapted generative models with innovations on sketch processing for sketch-based object and scene generation (2D and 3D) with spatial and semantic controls (ECCV '22). The model allows easy control and editing for users with sketches.
- **3D Object and Scene Modeling** : Developed label-free geometric and semantic representation learning; achieved state-of-the-art out-of-domain object recognition and pose estimation (ECCV Oral) and improved 3D point cloud segmentation for **autonomous driving** (TPAMI).
- **Long-Tailed and Open-World Recognition** : Tackled large-scale imbalanced learning for real-world applications including wildlife monitoring and ocular disease diagnosis (CVPR Oral).

WORK EXPERIENCE

Accenture, AI Platform & Research Center

Mountain View, CA

Senior ML Scientist and Tech Lead

Mar 2026 - Present

Senior ML Scientist

Sep 2025 - Mar 2026

- **Efficient and Unified Conditional Contextual Generation** : Led a visual-text [unified conditioning paradigm](#) that waives text encoders from diffusion models, cutting inference time by 59% at SOTA quality (NeurIPS '26 submission); enables spatially-grounded contextual image generation; extending to conditional video generation.
- **World Models for Spaces** : Lead an ongoing Stanford collaboration (Prof. [Jiajun Wu](#)) on agentic, physical-action-aware world models for interactive scene generation (ICLR target).
- **Memex – Scaling Long-Horizon LLM Agents** : Co-led [Memex](#), an indexed experience-memory mechanism that compresses agent working context without discarding evidence, with *Memex-RL* reward shaping it under a context budget – yielding higher long-horizon task success at substantially smaller working context. Implemented innovative features – Memex and multimodal self-evolving agents with DSL – into Accenture's mid-market agentic AI platform.

California Institute of Technology

Pasadena, CA

Postdoctoral Researcher, Computing and Mathematical Sciences

July 2023 - Aug. 2025

Mentor : [Anima Anandkumar](#)

- **Physics-Aware Generative AI** : Developed resolution-agnostic deep architectures with diffusion and feed-forward networks for unconditional and conditional image and 3D generation/reconstruction (CVPR '25). Developed neural simulator for ultrasound and photoacoustic physics (2D or 3D videos) using large-scale physics-aware neural networks (Nature submission).
- **Multi-Modal Foundation Models** : Developed a multi-modal foundation model (video, audio & text) that classifies different types of surgical feedback and its effectiveness in trainee behavior change, leveraging self-supervised fine-tuning techniques (2× ML4H Best Paper Awards).

Amazon

Sunnyvale, CA

Applied Scientist Intern

May 2022 - Nov. 2022

Student Researcher (via UC Berkeley)

May 2021 - May 2022

- **3D-Aware, Multi-View Virtual Try-On** : Developed multi-view human reposing and virtual try-on system that beats SOTA; shipped into *Amazon Virtual Try-On* product. [Patent](#) issued.
- **Sketch-Based Home/Scene Synthesis** : Developed an unsupervised scene-sketch-to-photo generative model for sketch-based home design (ECCV Workshop **Spotlight** '22, see [C15]).

Aizip

Cupertino, CA

Founding Member, Research Scientist

Oct 2020 - Aug. 2023

- Led core projects towards robust, efficient and scalable real-world **AI-IoT** solutions.
- **Full-stack machine learning** and delivered robust models and products to customers.
- Built the [tiniest human detection](#) system with robust performance under different lighting conditions.

Sensetime

Shenzhen, China

Research Scientist Intern

Feb. 2018 - Aug. 2018

- Built an RGBD-camera-based 3D portrait animation system—deployed in Vivo's flagship phone.
- Designed efficient classification algorithms for long-tailed fine-grained visual categorization datasets; ranked 6th globally in the [CVPR 2018 FGVC Challenge](#).

- CONFERENCE / WORKSHOP PAPERS
- [C1] [UniVL: Unified Vision-Language Embedding for Spatially Grounded Contextual Image Generation](#)
J. Wang, Y. Wang, W. Gan, Z. Wang, W. Wei
In submission to NeurIPS, 2026.
 - [C2] [Inference-Time Optimization with Confidence Dynamics](#)
Y. Wang, M. Liu, J. Wang, J. Huang, A. Shah, W. Wei
International Conference on Machine Learning (ICML), 2026
 - [C3] [Memex \(RL\): Scaling Long-Horizon LLM Agents via Indexed Experience Memory](#)
Z. Wang, H. Chen, J. Wang, W. Wei
arXiv :2603.04257, 2026
 - [C4] [Resolution-Independent Neural Operators for Multi-Rate Sparse-View CT](#)
A. Datta*, J. Wang*, A. Aali, A.S. Jatyani, A. Anandkumar
European Conference on Computer Vision (ECCV), 2026
 - [C5] [Coarse-to-Fine 3D MRI Reconstruction via Resolution-Agnostic Neural Operators](#)
J. Wang, R. Wang, V. Duruisseaux, R. Daftari, W. Gan, A.S. Jatyani, B. Bonev, A. Anandkumar
In submission to NeurIPS, 2026.
 - [C6] [Open-Vocabulary Monocular 3D Object Detection](#)
J. Yao, H. Gu, X. Chen, J. Wang, Z. Cheng
International Conference on 3D Vision (3DV), 2025
 - [C7] [A Unified Model for Compressed Sensing MRI Across Undersampling Patterns](#)
A. Jatyani*, J. Wang*, A. Chandrashekar, Z. Wu, B. Tolooshams, M. Liu-Schiaffini, A. Anandkumar
Conference on Computer Vision and Pattern Recognition (CVPR), 2025
 - [C8] [Pose-Aware Self-Supervised Learning with Viewpoint Trajectory Regularization](#)
J. Wang, Y. Chen, S.X. Yu
European Conference on Computer Vision (ECCV), ORAL (2%), 2024
 - [C9] [Insight: A Multi-Modal Diagnostic Pipeline using LLMs for Ocular Surface Disease Diagnosis](#)
C.H. Yeh, J. Wang, A.D. Graham, A. Liu, B. Tan, Y. Chen, Y. Ma, M.C. Lin
Conference on Medical Image Computing and Computer Assisted Intervention (MICCAI), 2024
 - [C10] [Multi-Modal Self-Supervised Learning for Surgical Feedback Effectiveness Assessment](#)
A. Gupta, R. Kocielnik, J. Wang, F. Nasriddinov, C. Yang, E. Wong, A. Anandkumar, A. Hung
Machine Learning for Health. PMLR. BEST PAPER, 2024
 - [C11] [Human Reposing and Virtual Try-On from Multi-View Images](#)
J. Wang, A. Kheradmand, H. Arora (*Accepted; withdrawn due to product conflicts.*)
Conference on Computer Vision and Pattern Recognition (CVPR), 2023
 - [C12] [Recurrent Parameter Generators](#)
J. Wang, Y. Chen, S.X. Yu, B. Cheung, Y. LeCun
Winter Conference on Applications of Computer Vision (WACV), 2023
 - [C13] [Deep Multimodal Fusion for Surgical Feedback Classification](#)
R. Kocielnik, E. Wong, T. Chu, L. Lin, D. Huang, J. Wang, A. Anandkumar, A. Hung
Machine Learning for Health. PMLR. BEST PAPER, 2023
 - [C14] [3D Shape Reconstruction from Free-Hand Sketches](#)
J. Wang, J. Lin, Q. Yu, R. Liu, Y. Chen, S.X. Yu
European Conference on Computer Vision (ECCV) Workshop, 2022
 - [C15] [Unsupervised Scene Sketch to Photo Synthesis](#) (Amazon collaboration; sketch-based home design)
J. Wang, S. Jeon, S.X. Yu, X. Zhang, H. Arora, Y. Lou
European Conference on Computer Vision (ECCV) Workshop, SPOTLIGHT, 2022
 - [C16] [Tracking the Dynamics of the Tear Film Lipid Layer](#)
T. Kothpalli, C. Shou, J. Ding, J. Wang, A.D. Graham, T. Svitova, S.X. Yu, M.C. Lin
Conference on Neural Information Processing Systems (NeurIPS) Workshop, 2022
 - [C17] [Orthogonal Convolutional Neural Networks](#)
J. Wang, Y. Chen, R. Chakraborty, S.X. Yu
Conference on Computer Vision and Pattern Recognition (CVPR), 2020
 - [C18] [Large-scale Long-Tailed Recognition in an Open World](#)
Z. Liu, Z. Miao, X. Zhan, J. Wang, B. Gong, S.X. Yu
Conference on Computer Vision and Pattern Recognition (CVPR), ORAL (5%), 2019
 - [C19] [Sur-Real: Frechet Mean and Distance Transform for Complex-Valued Deep Learning](#)
R. Chakraborty, J. Wang, S.X. Yu
Conference on Computer Vision and Pattern Recognition (CVPR) Workshop, BEST PAPER, 2019
 - [C20] [Point to Set Similarity Based Deep Feature Learning for Person Re-identification](#)
S. Zhou, J. Wang, J. Wang, Y. Gong, N. Zheng
Conference on Computer Vision and Pattern Recognition (CVPR), 2017

JOURNAL
ARTICLES

- [J1] [Self-Supervised Learning via Flow-Guided Neural Operator on Time-Series Data](#)
D. Nguyen*, J. Yao*, **J. Wang***, J. Berner, A. Anandkumar
Transactions on Machine Learning Research (TMLR), 2026
- [J2] [Beyond Closure Models: Learning Chaotic-Systems via Physics-Informed Neural Operators](#)
C. Wang, J. Berner, Z. Lin, D. Zhou, **J. Wang**, J. Bae, A. Anandkumar
In submission to Nature Communications, 2025
- [J3] [Ultrasound Lung Aeration Map via Physics-Aware Neural Operators](#)
J. Wang, O. Ostras, B. Tolooshams, M. Sode, Z. Li, K. Azizzadenesheli, G.F. Pinton, A. Anandkumar
In submission to Nature Biomedical Engineering, 2025
- [J4] [Variable Sampling for Fast and Efficient Functional Ultrasound Imaging using Neural Operators](#)
B. Tolooshams, L. Lin, T. Callier, **J. Wang**, et al.
In submission to Nature Methods, 2025
- [J5] [Artificial Intelligence Models Utilize Lifestyle Factors to Predict Dry Eye Related Outcomes](#)
A.D. Graham, **J. Wang**, T. Kothpalli, J. Ding, H. Tasho, A. Molina, V. Tse, S.M. Chang, S.X. Yu, M.C. Lin
Nature Scientific Reports, 2024
- [J6] [A Machine Learning Approach to Predicting Dry Eye-Related Signs, Symptoms and Diagnoses](#)
A.D. Graham, T. Kothpalli, **J. Wang**, J. Ding, V. Tse, P. Asbell, S.X. Yu, M.C. Lin
Heliyon, 2024
- [J7] [Open Long-Tailed Recognition in a Dynamic World](#)
Z. Liu, Z. Miao, X. Zhan, **J. Wang**, B. Gong, S.X. Yu
IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI), 2022
- [J8] [Predicting Demographics from Meibography Using Deep Learning](#)
J. Wang, A.D. Graham, S.X. Yu, M.C. Lin
Nature Scientific Reports, 2022
- [J9] [Spatial Transformer for 3D Point Clouds](#)
J. Wang, R. Chakraborty, S.X. Yu
IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI), 2021
- [J10] [Quantifying Meibomian Gland Morphology Using Artificial Intelligence](#)
J. Wang, S. Li, T.N. Yeh, R. Chakraborty, A.D. Graham, S.X. Yu, M.C. Lin
Optometry and Vision Science, 2021
- [J11] [A Deep Learning Approach for Meibomian Gland Atrophy Evaluation in Meibography Images](#)
J. Wang, T.N. Yeh, R. Chakraborty, S.X. Yu, M.C. Lin
Translational Vision Science and Technology (TVST), 2019
- [J12] [Insights and Approaches Using Deep Learning to Classify Wildlife](#)
Z. Miao, K.M. Gaynor, **J. Wang**, Z. Liu et al.
Nature Scientific Reports, 2019
- [J13] [Deep Ranking Model by Large Adaptive Margin Learning for Person Re-identification](#)
J. Wang, S. Zhou, J. Wang, Q. Hou
Pattern Recognition (PR), 2018

PATENT

- [1] [Image Generation Based on a Multi-Image Set and Pose Data](#)
A. Kheradmand, **J. Wang**, H. Arora
US Patent 12,354,337.

HONORS AND
AWARDS

Finalist, Perpall Speaking Competition, Caltech	2024
Best Paper Award, Machine Learning for Health (ML4H) Symposium	2023, 2024
Vector Institute Fellowship (offered)	2023
Best Paper Award, Hong Kong Science & Technology Park (HKSTP)	2019
Best Paper Award, CVPR PBVS workshop	2019
Seagate Fellowship	2018
Outstanding Graduate Award, Xi'an Jiaotong University	2018
Top 10 Undergraduate Award, Xi'an Jiaotong University	2017
National Scholarship of China	2015 - 2017
Meritorious Winner, the International Mathematical Contest in Modeling (top 8%)	2016